

Learning the rational choice perspective: A reinforcement learning approach to simulating offender behaviours in criminological agent-based models

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ABSTRACT

Over the past 15 years, environmental criminologists have explored the application of agent-based models (ABMs) of crime events and various theoretical frameworks applied to understand them. Models have supported criminological theorising and, in some cases, been applied to make predictions about the impact of interventions devised to reduce crime. However, decision-making frameworks utilised in criminological ABMs have typically been implemented through traditional techniques such as condition-action rules. While these models have provided significant insights, they neglect a crucial component of theoretical accounts of offending, the notion that offenders are learning agents whose behavioural dynamics change over time and space. In response, this article presents an ABM of residential burglary in which offender agents utilise reinforcement learning (RL) to learn behaviours. This solution enables offender agents to learn from individual-level perceptions of the environment and, given these perceptions, develop behavioural responses that benefit themselves. The model includes conceptualisations of the Routine Activity Theory (RAT), Crime Pattern Theory (CPT) and a utility function, Target Attractiveness, which acts as a behavioural mould to nudge offender agents to learn behaviours in keeping with the Rational Choice Perspective (RCP). Trained behaviours are then tested by introducing crime prevention interventions into the model and examining the reactions of offender agents. In keeping with empirical studies of offending, experimental results demonstrate that offender agents utilising RL learn to offend at targets where rewards outweigh risks and effort, offend close to home, frequently victimise high-rewarding targets, and conversely learn to avoid offending in areas associated with high levels of risk and effort.